



Marine Bird Ecology and Conservation, MSL F410 (CRN 36428)

3 Credits

Spring, 2023

Land Acknowledgement

- The University of Alaska Fairbanks' (UAF) Troth Yeddha' campus in Fairbanks, Alaska is located on the unceded, traditional homelands of the Lower Tanana Dene.
- UAF's College of Fisheries and Ocean Sciences campus in Juneau, Alaska is located on the unceded territories of the Áak'w Kwáan.
- A map of Indigenous Peoples and Languages of Alaska can be found [here](#).

COVID-19 statement

Students should keep up-to-date on the university's policies, practices, and mandates related to COVID-19 by regularly checking [this website](#). Further, students are expected to adhere to these policies, practices, and mandates and are subject to disciplinary actions if they do not comply.

General Information

<i>Instructor:</i>	Dr. Kristen Gorman	<i>Office Location:</i>	Irving II, room 230 (now) Irving II, room 231 (soon)
<i>Email:</i>	kbgorman@alaska.edu	<i>Office Hours:</i>	Mon and Wed 11:00-12:00, or anytime by appointment both in-person or via Zoom. Please email the instructor to set up an appointment.
<i>Telephone:</i>	907-474-5930	<i>*Course Type:</i>	Lecture: In-person, Online, Synchronous
<i>**Course Location:</i>	In-person: O'Neill 214 Online: https://alaska.zoom.us/j/mtqzliHdXWEXxUfnD8O6ANCTCmIVHf Canvas is our Learning Management System.	<i>Meeting Time:</i>	Tuesday/Thursday 09:45- 11:15 (Jan 17 – Apr 27, 2023) Final: Thursday May 4, 2023, 08:00-11:00



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Additional General Information

- As this course's instructor, I respectfully acknowledge Alaska's First Peoples and their long history of living in deep connection with their land and water, and the historical and ongoing legacy of colonialism. As a life-long student, I am committed to learning about injustice, and working to improve equity in my professional and personal life.
- This course meets in-person and online via Zoom during lecture hours (i.e., synchronously, see details above). **All students** (i.e., in-person or online attendees) must register in advance for the Zoom meeting to allow for the possibility of meeting entirely remotely due to COVID-19 or other unforeseen circumstances. After registering, you will receive a confirmation email containing information about joining the meeting, **you can save this information to your Google calendar to easily find the Zoom link each week.**
- Our in-class sessions will be recorded via Zoom to provide flexibility to those students who cannot attend. These recordings will be posted on Canvas (see weekly Modules), and therefore only available to enrolled students in this class. Despite classes being recorded, **students who will miss class must notify the instructor in advance and provide a valid reason** (see details below).
- If you need assistance navigating Zoom or Canvas please contact your instructor or [UAF's eCampus Student Support](#).

Prerequisites

BIOL F371 - Principles of Ecology, or MSL F320 - Aquatic Ecology, or by instructor approval.

Course Description

This course will introduce students to the biology, ecology, and conservation of marine birds with emphasis on seabirds, sea ducks, and shorebirds, especially species found in Alaska and the northern hemisphere. Through ecological and evolutionary perspectives, topics will include biodiversity, adaptations, life histories, population ecology, demography, community ecology, and conservation.

Additional Course Description

As we will be discussing broad themes in ecology and evolution through the lens of marine birds, lectures will also provide short primers on these broad themes, e.g., population genetics, phylogenetics, demography/vital rates, community assembly, as review prior to our discussion of specific details for our study species - seabirds, sea ducks, and shorebirds. Thus, the course should be viewed as both a review of these broad ecological and evolutionary themes that you should have/might have encountered in other coursework, as well as an introduction to the biology, ecology, physiology, population structure, and conservation of marine birds.



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Representative Course Readings/Materials

- All required reading materials will be provided.
- *Biology of Marine Birds*, E.A. Schreiber and J. Burger (editors). 2001. CRC Press. ISBN: 0849398827.
- *Ecology and Conservation of North American Sea Ducks*, J-P.L. Savard, D.V. Derksen, D. Esler, J.M. Eadie (editors). 2015. CRC Press. ISBN: 9781482298982.
- *Shorebird Ecology, Conservation, and Management*, M.A. Colwell. 2010. University of California Press. ISBN: 9780520266407.
- *Diving Physiology of Marine Mammals and Seabirds*, P.J. Ponganis. 2015. Cambridge University Press. ISBN: 9781139045490.
- *Conservation of Marine Birds*, L. Young, E. VanderWerf. 2022. Academic Press. ISBN: 9780323885393.

Technology Requirements

To get the most out of this course, it will be important to have regular access to a computer and the internet to view and download online materials in [Canvas](#). At various points during the semester, you will need to download course material and upload completed assignments. For the purposes of this course, **please use your @alaska.edu email account only for correspondence with the instructor**, or the Canvas inbox feature to ensure that you do not miss any important information from the instructor or from other university departments.

Course Goals

Marine birds are key predators of pelagic and coastal ecosystems (including inland waterways!), and many species (particularly seabirds) are among the most threatened vertebrates. With life histories closely linked to marine processes, i.e., ocean productivity, physical properties such as ocean current and upwelling, or specific dietary niches, the abundance, distribution, productivity, and overall health of marine bird populations are a window into the structure and function of the larger ecosystem. For these reasons, marine birds are important species for monitoring, management, and biodiversity conservation. Importantly, many species hold special roles in indigenous cultures and local communities for harvest, ritual, and eco-tourism. Further, seabirds are important organismal models in behavioral, evolutionary, and disease ecology, for these reasons, students of fisheries and marine science should have a solid command of the biology, ecology, and conservation of marine birds, which this course will provide.

Student Learning Outcomes

By the end of the course students will be able to:

- Describe the diversity of marine birds of the northern latitudes, and explain key biological and physiological adaptations unique to different taxonomic groups.



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- Understand the connections between the marine environment and species' life history strategies, population ecology, and demography.
- Apply your knowledge of marine bird systems to interpret conservation strategies in the context of environmental and ecological change.

Instructional Methods

The course is lecture based. You are expected to attend lectures, review specific reading material ahead of class as communicated by the instructor, and contribute to class discussions. Students who must attend remotely will be able to do so via Zoom link. Lecture slides will be made available after class and lectures will be recorded on request for students with excused absences or those attending asynchronously.

Explanation of Student Effort

Students are expected to attend 3 hours of lecture each week, and spend another 6 hours each week reading, preparing assignments and presentations, or studying for exams. In general, students should expect to spend about 10 hours each week on this course.

Tentative Course Calendar (subject to change following instructor notice)

Week	Lecture	Date	Topic	Readings	Delivery
1	1 – KBG	Tues, Jan 17	Course Introduction, Seabird biodiversity and distribution	Ch 1 & 3, Schreiber and Burger; additional readings TBA	Synchronous
1	2 – KBG	Thurs, Jan 19	Sea duck and shorebird biodiversity and distribution	Ch 2 & 13, Savard et al.; Ch 1 & 2, Colwell; additional readings TBA	Synchronous
2	3 – KBG	Tues, Jan 24	Biology and physiology of seabirds	Ch 11, 12, 13, & 14; Schreiber and Burger; additional readings TBA	Asynchronous
2	4 – KBG	Thurs, Jan 26	Biology and physiology of sea ducks	Ch 5 & 7, Savard et al.; additional readings TBA	Asynchronous
2		Fri, Jan 27	<i>Last day to add class or drop with refund</i>	-----	
3	5 – KBG	Tues, Jan 31	Biology and physiology of shorebirds	Ch 3, Colwell; additional readings TBA	Synchronous
3	6 – KBG	Thurs, Feb 2	Seabird life histories and adaptations	Ch 4, 8, & 10, Schreiber and Burger; Ch 2 & 6, Ponganis; additional readings TBA	Synchronous

3		Fri, Feb 3	Assignment 1 due by 23:59 AK time.		
4	7 – KBG	Tues, Feb 7	Sea duck life histories and adaptations	Ch 8, 9, 10, & 11 Savard et al.; additional readings TBA	Asynchronous
4	8 – KBG	Thurs, Feb 9	Shorebird life histories and adaptations	Ch 4, Colwell; additional readings TBA	Asynchronous
5		Tues, Feb 14	Mid-Term Exam 1		
5	9 – KBG	Thurs, Feb 16	Seabird population ecology I - annual cycle, wintering, breeding, and molting ecology	Ch 4, 8, & 9, Schreiber and Burger; additional readings TBA	Synchronous
6		Mon, Feb 20	Mid-term grades due	-----	
6	10 – KBG	Tues, Feb 21	Seabird population ecology II - foraging ecology, demography	Ch 6 & 16, Schreiber and Burge; additional readings TBA	Synchronous
6	11 – KBG	Thurs, Feb 23	Seabird population ecology III - predation, disease, other topics	Ch 15 Schreiber and Burger; Khan et al. 2019; additional readings TBA	Synchronous
7	12 – KBG	Tues, Feb 28	Sea duck population ecology I - annual cycle, wintering, breeding, and molting ecology	Ch 8, 9, 10 in Savard et al.; additional readings TBA	Synchronous
7	13 – KBG	Thurs, Mar 2	Sea duck population ecology II - foraging ecology, demography	Ch 7 Savard et al; additional readings TBA	Synchronous
7		Fri, Mar 3	Assignment 2 due by 23:59 AK time		
8	14 – KBG	Tues, Mar 7	Sea duck population ecology III - predation, disease, other topics	Ch 2, Savard et al.; additional readings TBA	Synchronous
8	15 – KBG	Thurs, Mar 9	Shorebird population ecology I - annual cycle, wintering, breeding, and molting ecology	Ch 5 & 6, Colwell; additional readings TBA	Synchronous
9		Tues, Mar 14	No class–Spring Break	-----	
9		Thurs, Mar 16	No class–Spring Break	-----	

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10	16 – KBG	Tues, Mar 21	Shorebird population ecology II and III - foraging ecology, demography, predation, disease, other topics	Ch 7, 8, & 9, Colwell; additional readings TBA	Synchronous
10		Thurs, Mar 23	Mid-Term Exam 2		
11	17 - KBG	Tues, Mar 28	Marine bird study methods and techniques I	Tremblay et al. 2009; Andrews and Enstipp 2016; additional readings TBA	Synchronous
11	18 – KBG	Thurs, Mar 30	Marine bird study methods and techniques II	Barrett et al. 2007; Hollmen et al. 2002; additional readings TBA	Synchronous
11		Fri, Mar 31	Assignment 3 due by 23:59 AK time		
11		Fri, Mar 31	Deadline for student- and faculty-initiated withdrawals (W on transcript)	-----	
12	19 – KBG	Tues, Apr 4	Seabird community ecology	Ballance et al. 1997; additional readings TBA	Synchronous
12	20 – KBG	Thurs, Apr 6	Sea duck community ecology	Lamb et al. 2019; additional readings TBA	Synchronous
13	21 – KBG	Tues, Apr 11	Shorebird community ecology	Kuwae et al. 2012; additional readings TBA	Synchronous
13	22 – KBG	Thurs, Apr 13	Conservation and management of seabirds	Wagner and Boersma 2011; Oro 2014; Clairbaux et al. 2019; additional readings TBA	Synchronous
14	23 – KBG	Tues, Apr 18	Conservation and management of sea ducks	Ch 14 Savard et al; Koneff et al. 2017; additional readings TBA	Synchronous
14	24- KBG	Thurs, Apr 20	Conservation and management of shorebirds	Ch 11, Colwell; additional readings TBA	Synchronous
14		Fri, Apr 21	Assignment 4 due by 23:59 AK time		
15	25	Tues, Apr 25	Student Presentations		

15	26	Thurs, Apr 27	Student Presentations & Review for Final		
16	27	Thurs, May 4	FINAL EXAM	08:00-10:00	

Assignments, Activities, and Evaluation

- The course is primarily lecture- and discussion-focused during meeting times. Thus, your attendance and participation in class meetings is required and will be calculated as part of your grade. Participation is key to student learning as it fosters communication and peer to peer discussion. For example, we encourage at least one comment or question per class period by each student. Students will earn 2 points for each class attended (50 points), and up to 50 points for contributing to class discussions over the entire semester. A total of 100 points towards your final grade.
- There will be four assignments throughout the semester covering major themes from the course, details will be posted in Canvas. Each assignment is worth 25 points. Total 100 points towards your final grade.
- There will be two mid-semester exams and one final exam each semester (100 points each). Exams will be written, short-answer, and multiple-choice questions. The final exam will include material presented throughout the semester, but will focus on material covered after the second exam (100 points). A total of 300 points towards your final grade.
- You will be responsible for a 15-minute presentation on a topic of your choice. Student presentations should build on course materials and focus on a specific area of marine bird science of interest to the student. A total of 100 points towards your final grade.
- Please review the course grading scheme below for the weighting of each activity in your final grade calculation.

Grading

	Possible points	% of total grade
Midterm Exam #1	100	15%
Midterm Exam #2	100	15%
Final Exam	100	20%
Assignments (4)	100 (25 each)	20%
Student Presentation (15 minutes)	100	20%
Attendance and Participation	100	10%
Total	600	100%



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Course grades will be assigned according to the following scale:

A = 90-100 %

B = 80-89 %

C = 70-79 %

D = 69-60 %

F = less than 60%

Your instructor follows the University of Alaska Fairbanks Incomplete Grade Policy:

"The letter "I" (Incomplete) is a temporary grade used to indicate that the student has satisfactorily completed (C- or better) the majority of work in a course but for personal reasons beyond the student's control, such as sickness, has not been able to complete the course during the regular semester.

Negligence or indifference are not acceptable reasons for an "I" grade." For more information, see [the UAF regulations regarding grades](#).

Course Policies

- As with all members of the UAF community, the university requires all in-person and online students to conduct themselves honestly and responsibly and to respect the rights of others. Students may not engage in behavior that disrupts the learning environment, violates the rights of others, or otherwise violates the student code of conduct, university rules, regulations, or procedures. More information about the UAF Student Code of Conduct and Student Rights and Responsibilities can be found [here](#).
- High ethical standards are expected of all students in this course. Violations of the UAF Student Code of Conduct will result in an immediate "F" grade. Additionally, violations of the Student Code of Conduct may result in disciplinary action initiated by UAF.
- Students can miss one class unexcused, but reviewing the material covered during the missed period is your responsibility. The instructor strongly encourages discussions, questions, and noting any topics of interest in popular media during class. Come to class ready to engage with the course material, your fellow classmates, and the instructors.
- Extended absence policy: The University of Alaska Fairbanks recognizes that students may need to miss more classes than are allowed by a particular instructor as specified in course policies. Extended absences are defined as missed classes or course work by students beyond what is permissible by the instructor's written course policies. Students may need to miss class and/or course work for a variety of reasons, including, but not limited to: Bereavement, Personal illness or injury, Serious illness of a friend, family member or loved one, Military obligations, Jury service, Other emergency or obligatory situations. Please speak with your instructor concerning any situation that might require an extended absence. For more information, go to the Students Handbook or the [Center for Students Rights and Responsibilities](#).



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- Please make sure you use your current @alaska.edu email address for all course correspondence. Check your e-mail messages regularly. Class information and changes to the syllabus will be distributed through Canvas to your email. Please note that the course schedule is “tentative”, and the “actual” course schedule will be a living document that evolves over the semester (see Canvas).
- This course will require students to have access to the internet, a personal computer, and familiarity with basic computer software programs such as Word, PowerPoint, and Excel.
- Make-up exams will generally not be allowed unless there are extenuating circumstances that are communicated in advance of the exam. Plan to take the exams during the scheduled class time. If you miss the exam due to an unexcused absence you will receive zero points.
- For student presentations, species or topics cannot be covered multiple times by different students, thus contact your instructor to ensure that you can present on the species/topic of your choice. Presentation scheduling is also determined on a first-come-first-serve basis.
- Cheating, Forgery, Plagiarism, Disruptive Behavior, etc. High ethical standards are expected of all students. The standards in this class adhere to the UAF Student Code of Conduct that can be found here: <https://uaf.edu/deanofstudents/student-code-of-conduct/disciplinary-processes/student-code-of-conduct/>. Violations of the UAF Student Code of Conduct will result in an immediate “F” grade. Additionally, violations of the Code of Conduct may result in disciplinary action initiated by UAF.

A note on plagiarism. Defined as “A piece of writing that has been copied from someone else and is presented as being your own work”. This means that information from books, scientific literature, internet, even images, etc, must be properly cited. However, there also needs to be some original thought in your work – you cannot take an entire presentation or even a whole slide, reference it as “XYZ” and be done with it. When in doubt, ask your instructor.

Support Services

- UAF libraries are a great resource. While on the UAF network, many journals recognize the university authorization to access scientific papers so you can download and print papers of interest to you, your studies, or this class. If a paper is not available, you can request them through Interlibrary Loan: <http://library.uaf.edu/ill>
- Google Scholar <http://scholar.google.com/schhp?hl=en> is one of the greatest tools to search for scientific literature, images, and videos based on keywords. If you are on campus you can immediately link to pdf files of most papers.
- UAF Student Support Services <http://www.uaf.edu/sssp/> offers general support ranging from note taking to career planning.



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Student Protections Statement

UAF embraces and grows a culture of respect, diversity, inclusion, and caring. Students at this university are protected against sexual harassment and discrimination (Title IX). Faculty members are designated as responsible employees which means they are required to report sexual misconduct. Graduate teaching assistants do not share the same reporting obligations. For more information on your rights as a student and the resources available to you to resolve problems, please go to the following site:

<https://catalog.uaf.edu/academics-regulations/students-rights-responsibilities/>.

Disabilities Services Statement

I will work with the Office of Disability Services to provide reasonable accommodation to students with disabilities. The Office of Disability Services implements the Americans with Disabilities Act (ADA), and ensures that UAF students have equal access to the campus and course materials. I will work with the Office of Disabilities Services (208 Whitaker, 907-474-5655) to provide reasonable accommodation to students with disabilities uaf.edu/disability/

Non-Discrimination Statement

The University of Alaska is an affirmative action/equal opportunity employer and educational institution. The University of Alaska does not discriminate on the basis of race, religion, color, national origin, citizenship, age, sex, physical or mental disability, status as a protected veteran, marital status, changes in marital status, pregnancy, childbirth or related medical conditions, parenthood, sexual orientation, gender identity, political affiliation or belief, genetic information, or other legally protected status. The University's commitment to nondiscrimination, including against sex discrimination, applies to students, employees, and applicants for admission and employment. Contact information, applicable laws, and complaint procedures are included on UA's statement of nondiscrimination available at www.alaska.edu/nondiscrimination. For more information, contact:

UAF Department of Equity and Compliance
1760 Tanana Loop, 355 Duckering Building, Fairbanks, AK 99775
907-474-7300
uaf-deo@alaska.edu

Student Academic Support

- **[Speaking Center](#)**
 - 907-474-5470
 - uaf-speakingcenter@alaska.edu
 - Gruening 507
- **[Writing Center](#)**
 - 907-474-5314



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- uaf-writing-center@alaska.edu
- Gruening 8th floor
- **[UAF Math Services](#)** (math fee paying students only)
- uafmathstatlab@gmail.com
- Chapman Building
- **[Developmental Math Lab](#)**
- CTC, Room 120, 604 Barnette Street
- Gruening 406
- **[The Debbie Moses Learning Center](#)** at CTC
- 907-455-2860
- 604 Barnette St, Room 120
- **[Elmer E. Rasmuson Library](#)** (help with research)
- 907-474-7481 (phone)
- 907-341-4404 (text)
- AskRasmusonLibrary@uaf.libanswers.com
- 1732 Tanana Loop
- **[Rural Student Services](#)**
- uaf-rss@alaska.edu
- [Tutoring Services](#)
- Main floor Brooks Building
- **[Student Support Services](#)**
- (907) 474-6844
- **[eCampus Student Services](#)**
- 907-479-3444
- toll free 1-800-277-8060
- [contact staff directly](#)
- **[Veterans Resources Center](#)**
- 907-474-2475
- Rm 111, Eielson Building.
- **[Academic Advising Resource List](#)**

Other Student Resources:

- **[Disability Services](#)**
- 907-474-5655
- uaf-disability-services@alaska.edu
- Whitaker 208
- **[Student Health & Counseling](#)** [6 free counseling sessions]
- 907-474-7043
- [Appointments](#)
- Whitaker 203
- **[Center for Student Rights and Responsibilities](#)**
- 907-474-7317
- uaf-studentrights@alaska.edu



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- Eielson 110
- **ASUAF Student Government**
- 907-474-7355
- asuaf.office@alaska.edu
- Wood Center 119

University Sponsored Off-Campus Programs and Research Activities

We want you to know that:

1. UA is an AA/EO employer and educational institution and prohibits illegal discrimination against any individual: www.alaska.edu/nondiscrimination.
2. Incidents can be reported to your university's Equity and Compliance office (listed below) or online reporting portal. University of Alaska takes immediate, effective, and appropriate action to respond to reported acts of discrimination and harassment.
3. There are supportive measures available to individuals that may have experienced discrimination.
4. University of Alaska's Board of Regents' Policy & University Regulations (UA BoR P&R) 01.02.020 Nondiscrimination and 01.04 Sex and Gender-Based Discrimination Under Title IX, go to: <http://alaska.edu/bor/policy-regulations/>.
5. UA BoR P&R apply at all university owned or operated sites, university sanctioned events, clinical sites and during all academic or research related travel that are university sponsored.

For further information on your rights and resources [click here](#).

Title IX

University of Alaska Board of Regents have clearly stated in BOR Policy that discrimination, harassment and violence will not be tolerated on any campus of the University of Alaska. If you believe you are experiencing discrimination or any form of harassment including sexual harassment/misconduct/assault, you are encouraged to report that behavior. If you report to a faculty member or any university employee, they must notify the UAF Title IX Coordinator about the basic facts of the incident.

Your choices for reporting include:

1. You may access confidential counseling by contacting the UAF Health & Counseling Center at 907-474-7043;
2. You may access support and file a Title IX report by contacting the UAF Title IX Coordinator at 907-474-6600;
3. You may file a criminal complaint by contacting the University Police Department at 907-474-7721. For more information please visit the [UAF Department of Equity and Compliance](#) webpage.



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Any UAF employee or volunteer who reasonably suspects or observes minor abuse or maltreatment is required to report the incident. Reporting procedures are available on the UAF Protection of Minors. Violation of this policy by employees shall be reported as well.

Emergency Notification Plan

Students will receive emergency notifications via phone or email. Please check your uaonline account to confirm your emergency notification settings. for more information, please refer to the student handbook. In cases where you do not have access to your devices, as your instructor, I will take responsibility to relay any emergency notifications.